

Example Library: First Principles-Driven Hybrid Models for Reaction Furnace Oxidation in Aspen HYSYS

Products: Aspen Hybrid Models for HYSYS

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Example Details

The AI Training feature in Aspen HYSYS enables you to use data and machine learning techniques to improve the accuracy of one or more models; used to compensate for deficiencies in the model. A neural network embedded within the model is trained using data to predict one or more model parameters as a function of feed and operating conditions.

We highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model



The attached PDF shows the steps to follow to ‘train’ a model with plant data using First Principles-Driven Hybrid Models in Aspen HYSYS to help us optimize the oxidation of H₂S in the reaction furnace at a high temperature.

Keywords

First Principles, Aspen HYSYS, SRU, Reaction furnace, H₂S Oxidation